

OPEN BOOK EXTRACTS x FACLON LABS

Operations & Data Integration

Current-State Pre-Read

Roxboro, North Carolina | September 2026

PURPOSE OF THIS PRE-READ

Provide Faclon's technical team with a concise, self-contained view of OBX's current production workflows, data systems, controlled records, and highest-value integration opportunities so the onsite discussion can focus on validation, architecture, and pilot definition rather than basic process discovery.

CURRENT-STATE FRAMING

Bulk cannabinoid production is currently paper-log driven. Kratom/MIT has a substantially stronger electronic genealogy and production-logging foundation, while important equipment and process telemetry remains local, manually entered, or not historically retained.

Prepared for Faclon Labs technical review

1. Executive Overview

OBX operates two manufacturing areas at different levels of digital maturity. Both have established production procedures and material-traceability practices; the shared opportunity is to create a single, time-aligned record that connects material genealogy, equipment conditions, operator actions, quality results, and final process outcomes.

The near-term objective is to establish a trusted data foundation before pursuing advanced AI or closed-loop process control. A successful first phase should improve batch reconstruction, reduce duplicate transcription, capture the variables most likely to explain yield and cycle-time variation, and preserve OBX's existing Quality and system-of-record requirements.

Dimension	Kratom / MIT	Bulk Cannabinoids
Process model	Continuous-batch extraction with batch downstream processing	Batch distillation, crystallization, and chemical conversion
Primary production record	Custom electronic applications with some manual / paper steps	Controlled paper BPRs and equipment logs
Material genealogy	Structured from biomass lot through tote, batch, drum, and salt run	Maintained through controlled records and inventory systems, but not centrally queryable
Equipment control	Beckhoff PLC upstream plus local controllers downstream	Primarily local controllers, VFDs, gauges, chillers, and equipment displays
Time-series process history	Limited relative to available PLC / instrument data	Minimal; most values are recorded as periodic paper observations
Primary opportunity	Connect existing telemetry to the current digital genealogy	Digitize controlled records and connect high-value equipment signals

SHARED DESIGN PRINCIPLE

Integrate with the systems and controlled workflows OBX already owns. The first objective is not "AI everywhere"; it is reliable, contextualized production data that can support analytics, SPC, and AI as the dataset matures.

2. Site Visit Objectives

The onsite discussion will be most valuable if it converts the current-state context into a practical implementation path. OBX would like the visit to establish the following outputs:

1. A validated current-state instrumentation and data architecture: what exists today, what can be connected with configuration only, and where new hardware is justified.
2. A narrowly scoped Phase 1 pilot with a defined business outcome, measurable success criteria, and clear technical dependencies.
3. A phased path from connectivity and electronic records to analytics / SPC and, later, AI-assisted optimization.
4. A practical ownership model covering Faclon scope, OBX scope, hardware, IT/OT integration, Quality involvement, and ongoing support.
5. Clear integration boundaries so existing OBX systems are extended rather than unnecessarily replaced.

PILOT SELECTION QUESTION

What is the smallest deployment that creates enough measurable operational value to justify building the broader data foundation?

3. Kratom / Mitragynine Current State

The Kratom/MIT operation already has a meaningful digital backbone. The strongest existing asset is structured material genealogy across biomass lots, extraction outputs, crystallization batches, freebase drums, salt conversion runs, and downstream transactions. The main limitation is that many physical process variables are manually entered, visible only on local instruments, or generated by the PLC without being retained as a complete historical record.

Biomass lot -> pretreatment / extraction -> extraction tote -> crystallization batch -> freebase drum -> acetate salt run -> QC / inventory

Existing digital foundation

Layer	Current System	Current Role
Operator applications	Browser / tablet production applications	Structured event entry, batch progression, genealogy
Application hosting	Google Cloud Run	Production web services / APIs
Operational database	Google Cloud SQL - PostgreSQL	Structured production genealogy and transactions
Supporting files	Google Cloud Storage	Selected production images / supporting records
Outside process control	Beckhoff PLC + HMI	Controls pumps, valves, tanks, and extraction logic; complete history is not centrally retained
Enterprise inventory	Acumatica	Inventory / accounting; selected operational handoffs remain manual
Quality data	In-house HPLC + external COAs	Quality results exist; automated linkage and completeness should be validated

Highest-value validation areas

Area	Data of Interest	Validation Focus
Outside extraction / PLC	Tank level, pump and valve state, cycle / drain timing, available flow and pressure tags	Confirm PLC / IPC model, TwinCAT version, tag availability, communications options, and historical buffering.
Solvent / LLE system	Heptane movement, organic / aqueous flow, tank level, phase / decant behavior	Identify existing metering and the minimum new instrumentation required for a credible solvent mass balance.
Crystallization	pH trajectory, internal / jacket temperature, base dosing rate and total, hold time	Identify instrument outputs and determine which signal best represents actual product conditions.
Scales / packaging	Gross, tare, net weight and material identity	Assess direct digital weight capture and barcode / QR confirmation while preserving existing genealogy.
Quality integration	Biomass, in-process, freebase, and salt assay results	Confirm how lab results and COAs can be linked back to process history without duplicating systems.

4. Bulk Cannabinoid Current State

The bulk cannabinoid area is well suited to a Foundation-stage digitization effort because the operating workflows are already structured and controlled on paper. The opportunity is not to redesign the manufacturing process; it is to convert the existing record structure into electronic batch records and selectively replace handwritten equipment observations with direct data capture where the installed instrumentation supports it.

Primary flow: Crude / WDC -> 1M WFE deterp -> 2M WFE distillation / FSD -> CBD isolate or derivative reaction / wash -> final purification / drying -> QC / inventory

RECOMMENDED EBR APPROACH

Use a hybrid electronic batch record: automatically populate equipment values and calculations where practical; retain explicit operator actions for material-condition observations, critical additions, exception handling, and Quality-required attestations.

Controlled production records reviewed

Record	Title	Current Revision	What It Captures
FOR-BIP-009	1M Distillation Log	Rev. 2 - 17NOV2025	Shift handoff; batch / mass / timing data; scheduled two-hour deterp parameter checks
FOR-BIP-010	2M Distillation Log	Rev. 2 - 17NOV2025	Shift handoff; oil-added confirmation; batch / mass / timing data; scheduled two-hour distillation parameter checks
FOR-BIP-004	Isolation Log	Rev. 7 - Effective 16JAN2026	Reactor A-E; up to four crashes; input mass; heptane; temperature/time milestones; transfer; rinses; fluff/mix; tray timing; initials / notes
FOR-BIP-003	D8 Log	Rev. 4 - Effective 05FEB2026	Reaction identity; input mass; heptane; material-in / acid-in times; exotherm temperature/time; set temperature/time; three wash checks

What the records and SOPs indicate

Workflow	Current-State Observation	Design Implication
Distillation - 1M / 2M	The current logs require scheduled snapshots of wiper, feed, run-off, target, vacuum, evaporator, condenser, and chiller set / actual values.	Determine which VFDs, controllers, and gauges can be read continuously. Preserve the scheduled human review / initials as an acknowledgement rather than requiring duplicate transcription.
CBD isolation	FOR-BIP-004 records cooling milestones, rinses, fluff / mix, and tray timing. SOP-BIP-003 defines an objective crystallization endpoint based on internal reactor temperature reaching at least -18 C while retaining visual material-condition checks.	Capture the actual internal temperature history and automatic milestone timestamps. Keep visual color / dryness / material-condition judgments as operator actions where the SOP requires them.
Delta-8 reaction	FOR-BIP-003 is a compact production summary. SOP-BIP-005 contains additional recipe and sequence logic, including PTSA and heptane calculations, a 50-55 C pre-acid condition, exotherm monitoring, an 85-95 C post-exotherm cook, and three defined washes.	Use input mass to drive recipe calculations and guided steps. Capture the temperature curve continuously so peak, rise, time-to-peak, and cook compliance can be derived rather than manually reconstructed.
Weights / solvents	Weights and solvent quantities are repeatedly transcribed by hand across the bulk workflows.	Assess connected scales first; add flow measurement only where it materially improves recipe control, solvent accounting, or investigation capability.

DOCUMENT-CONTROL ITEM

SOP-BIP-004 and SOP-BIP-006 both address distillation operations. Before any limits or recipes are encoded digitally, OBX Quality will confirm the current scope and governing-document relationship.

5. Data & Instrumentation Priorities

The table below consolidates the most important equipment and data targets for discovery. “TBD onsite” is intentional where make/model, available communications, or signal ownership has not yet been confirmed. The goal is to validate what can be connected before recommending new instrumentation.

Area / Asset	High-Value Data	Current State	Onsite Validation Focus
Kratom - Beckhoff PLC / IPC	Tank levels; valve / pump states; extraction and drain timing	Control data available; complete history not centrally retained	Confirm tag list, protocol options, network access, and edge buffering.
Kratom - LLE / solvent system	Organic / aqueous flow; levels; pump state; solvent return / loss	Incomplete historical mass balance	Inventory existing meters and define minimum new metering for a defensible heptane balance.
Kratom - crystallization	pH, product / jacket temperature, base dosing profile	Endpoints exist; continuous history incomplete	Identify pH, temperature, chiller, and dosing-pump interfaces; select the signal that best represents product conditions.
Kratom - scales / drumming / salt	Gross / tare / net; source identity; salt conversion weights	Weights largely typed into applications	Identify digital scale outputs and barcode / QR opportunities.
Bulk - 1M / 2M WFE	VFD values; vacuum; temperature set / actual; chiller values; mass outputs	Two-hour paper snapshots	Identify controller / VFD / gauge protocols and distinguish recipe / process-mode limits.
Bulk - ISO-A to ISO-E	Input mass; heptane charge; internal / jacket temperature; cooling milestones	Paper milestones + local displays	Identify true internal product-temperature source and practical heptane measurement point.
Bulk - RXN-1 / RXN-3	Temperature curve; heater setpoint; input mass; critical additions; reaction timing	Paper summary + local display	Determine high-frequency temperature capture path and whether addition quantities can be measured or electronically confirmed.
Bulk - Buchner / wash / drying	Rinse events; vacuum; filtration duration; oven / drying temperature and vacuum	Paper / local display	Capture objective timing / equipment data; preserve operator visual decisions where they remain process requirements.
Both - Acumatica / GCP / lab	Batch / lot identity; material transactions; assays / COAs; timestamps	Separate systems with manual handoffs	Define read / write boundaries, system-of-record ownership, API options, and common identity model.

WALKDOWN CLASSIFICATION

For each high-value variable, classify it as: (1) already structured electronically, (2) manually entered electronically, (3) measured / displayed but not retained, or (4) not currently measured. This separates digitization work from new instrumentation requirements.

6. Digital Architecture & Integration Boundaries

OBX's current-state architecture is intentionally mixed: Kratom/MIT has custom cloud applications and a relational production database, while the bulk cannabinoid area relies on controlled paper records and local equipment. The preferred future architecture should extend the existing systems rather than create parallel batch identities or competing systems of record.

Physical assets + operator actions -> edge connectivity / buffering -> time-series process history + eBPR -> existing OBX genealogy / quality / ERP systems -> analytics / SPC -> AI-assisted optimization

Layer	Applies To	Current System	Integration Principle
Production applications	Kratom / MIT	Google Cloud Run	Retain; expose appropriate APIs / event interfaces.
Operational database	Kratom / MIT	Google Cloud SQL - PostgreSQL	Retain as a core production genealogy source unless architecture review identifies a specific reason to change.
PLC / controls	Kratom upstream	Beckhoff PLC + HMI	Read process data without compromising control reliability; use appropriate OT segmentation and edge buffering.
Bulk production records	Bulk cannabinoids	Controlled paper BPRs / equipment logs	Digitize through Quality-controlled templates, audit trails, permissions, and electronic approvals.
ERP	Both	Acumatica	Define system-of-record ownership and the minimum read / write scope needed for Phase 1.
Quality results	Both	In-house HPLC + external COAs	Link results to existing batch identities; avoid a duplicate quality-data model where possible.
Collaboration	Both	Slack	Useful for notifications, but should not be the long-term transaction or integration backbone.

Architecture principles for discovery

- Preserve existing OBX batch and lot identities; avoid introducing a parallel genealogy model.
- Use least-privilege, read-only access for discovery where practical. Credentials, passwords, API secrets, and unrestricted database exports are not part of this pre-read.
- Design for temporary network or internet loss on the outdoor pad through appropriate edge buffering / local persistence.
- Keep process-control authority separated from analytics and data collection unless a future, validated control use case is intentionally approved.
- Treat Quality requirements for template control, audit trail, corrections, approvals, retention, and electronic signatures as design inputs to any bulk eBPR.

7. Initial Opportunity Areas

The following priorities are intended as a starting point for joint evaluation, not a fixed implementation sequence. The site visit should refine the ranking based on connection feasibility, implementation effort, business impact, and data dependencies.

Priority	Opportunity	Area	Rationale
1	Foundation connectivity + common batch context	Both	Connect available signals and align them to existing material / batch identities before adding unnecessary sensors.
2	Bulk electronic batch records	Bulk	Reduce transcription and review burden while preserving controlled records, approvals, and operator accountability.
3	Kratom PLC / process telemetry ingestion	Kratom	Add physical process history to an already-structured digital genealogy.
4	Heptane solvent mass balance	Kratom	Create a reliable solvent-cost / loss KPI and localize losses to the relevant unit operation.
5	Kratom crystallization dataset	Kratom	Build time-aligned pH, temperature, dosing, hold, filtration, yield, and quality history for future golden-batch analysis.
6	WFE continuous operating history	Bulk	Replace sparse paper snapshots with true operating profiles that can support deviation and yield analysis.
7	Connected scales + quality linkage	Both	Reduce weight transcription errors and connect process history to assay / release outcomes.
8	Advanced AI / predictive optimization	Both	Defer until enough trusted, contextualized data exists to train and validate useful models.

PILOT SUCCESS METRICS

A Phase 1 pilot should have a measurable operational KPI - for example: labor hours removed, record-review time reduced, solvent loss quantified / reduced, cycle time reduced, yield variance explained, genealogy exceptions reduced, or downtime avoided. "More data collected" should not be the sole success criterion.

Business questions the connected data foundation should enable

- Why did one lot or batch materially outperform another on yield, cycle time, or quality?
- Where is heptane actually being lost, and what is solvent loss per kilogram of finished MIT?
- Which pH, temperature, dosing, and hold profiles are associated with the best Kratom crystallization outcomes?
- Which WFE operating conditions are associated with yield or quality variation?
- Which assets or process steps are driving the most downtime or lost production time?
- Can a released batch be reconstructed end-to-end with its material lineage, operator events, equipment conditions, and quality results?
- Where is duplicate transcription creating avoidable labor, delay, or reconciliation risk?

8. Items to Validate Onsite

The points below reflect OBX's current understanding from the supplied records and internal system context. They are included so the onsite team can confirm the details that materially affect architecture and scope.

Topic	Current OBX View	Validate / Confirm
Kratom lot scale	Internal process documentation references typical truck / lot sizes of roughly 14,000-17,000 kg.	Confirm the current operating range and whether any recent sourcing / lot conventions differ.
Kratom upstream logging	Tablet applications and Beckhoff control both contribute to the current record.	Confirm which tank, cycle, drain, and transfer events are operator-entered versus directly available from PLC logic.
Kratom crystallization record	Structured batch / genealogy fields exist; continuous pH, temperature, and dosing history is not yet a complete central record.	Confirm the exact instrument sources and which milestones remain paper / manual.
Bulk production records	Current floor record is controlled paper.	Review a representative completed record for each major workflow to see real use of notes, corrections, initials, exceptions, and handoffs.
Solvent recovery	A reliable unit-operation solvent mass balance is not currently available.	Confirm what tank inventory, flow, or transfer measurements already exist and where additional measurement is economically justified.
Cloud SQL scope	Current Cloud SQL production data applies primarily to Kratom / MIT.	Confirm the cleanest integration pattern for future bulk eBPR and historian data.
Acumatica integration	Acumatica remains the enterprise inventory / accounting system.	Define read / write boundaries, permissions, and whether Phase 1 requires ERP write-back.
Distillation SOP hierarchy	SOP-BIP-004 and SOP-BIP-006 both address distillation.	OBX Quality to confirm governing scope before any digital recipe limits or alerts are configured.
Quality / electronic records	Bulk eBPR will require controlled templates, auditability, correction history, roles / permissions, and approval logic.	Confirm specific ISO / cGMP expectations and whether future customer requirements may justify 21 CFR Part 11 alignment.
Network / edge architecture	Outdoor extraction control and plant IT are separate operational concerns.	Confirm network segmentation, available outbound paths, edge location, local buffering, and cybersecurity requirements.

Technical discussion questions

1. Which existing instruments can be connected without interrupting production or materially changing validated operating practices?
2. What would a Faclon edge layer look like on the outdoor pad, and how would it buffer data during network interruptions?
3. How would Faclon represent OBX's existing batch genealogy so new process history attaches to the current identities rather than creating parallel records?
4. For bulk eBPR, how are controlled template revisions, audit trails, reason-for-change, electronic approvals, and record retention handled?
5. What data remains fully portable if OBX later changes vendors or develops additional capabilities internally?
6. Which comparable Faclon deployment is closest to a lean specialty-chemical / botanical batch operation rather than a large continuous plant?
7. What is the smallest Phase 1 scope Faclon would recommend after the walkdown, and what would OBX need to provide for a successful 30-60-90 day implementation?